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MILITARY SPECIFICATION

DUMMY WARHEAD, BOMB MDU-9/E

1. SCOPE. This specification covers one type of equipment designated Dummy Warhead, Bomb MDU-9/E, hereafter referred to as dummy warhead. This dummy warhead permits the training of personnel in handling and loading operation of the parent weapon.

2. APPLICABLE DOCUMENTS. The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

TT-T-266	Thinner: Dope and Lacquer (Cellulose-Nitrate)
TT-A-468	Aluminum-Pigment; Powder and Paste, For Paint

Military

JAN-P-100	Packaging and Packing for Overseas Shipment, General Specification For
MIL-C-104	Crates, Wood, Lumber and Plywood, Sheathed, Nailed and Bolted
MIL-P-116	Preservation, Methods Of
MIL-C-3774	Crate, Open, Wood
MIL-E-4970	Environmental Testing, Ground Support Equipment, General Specification For
MIL-L-7178	Lacquer, Cellulose Nitrate, Gloss, For Aircraft Use
MIL-T-9107	Test Reports, Preparation Of
MIL-L-10547	Liners, Case, Waterproof
MIL-B-27000	Bomb, Practice, General Specification For
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property

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PUBLICATIONS

Air Force Specifications Bulletin

No. F49	Codes for Ground Equipment in Finishing Schemes

DRAWINGS

Air Force

61J14300	Dummy Warhead, Bomb MDU-9/E

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 PREPRODUCTION. This specification makes provisions for preproduction testing.

3.2 COMPONENTS. The dummy warhead shall consist of the following major components:

a. Main body

b. Tail cover

c. Dummy timer

d. Arm/safe switch

e. Pressure valve

f. Electrical receptacles

3.3 GENERAL SPECIFICATION. The requirements of MIL-B-27000 apply as requirements of this specification with the exceptions and additions specified herein. When the two specifications conflict, this specification shall govern.

3.4 DESIGN AND CONSTRUCTION. Design and construction of the dummy warhead shall be in conformance to drawing 61J14300 and subsidiary drawings.

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3.4.1 DIMENSIONS. The dimensions and tolerances of the dummy warhead shall be:

Length	101.270 ± 0.165 inches
Diameter	Max. 37.000 ± 0.060 inches

3.4.2 INTERCHANGEABILITY. Parts functionally and dimensionally interchangeable shall be identified by the same part number. When parts are not interchangeable the criteria listed in MIL-D-70327 shall apply and new drawings shall be made as required.

3.4.3 WEIGHT. The weight of the dummy warhead shall be as specified on drawing 61J14300.

3.4.4 FINISH. The dummy warhead shall be finished as specified in MIL-B-27000. All steel surfaces shall be primed per USAF Specification Bulletin F49-12.

3.4.4.1 FINAL FINISH COAT. The external parts shall have a final finish of clear glossy lacquer per MIL-L-7178, colored with aluminum pigment paste per TT-A-468, type 2, class A. The lacquer shall be thinned by one volume of lacquer thinner per TT-T-266, to two volumes of lacquer. The lacquer shall be sprayed and a coating of 0.5 to 1.0 mil thickness shall be applied. All parts receiving final finish shall be allowed to air dry one hour before handling.

3.5 PERFORMANCE

3.5.1 FUNCTIONAL. Functional performance of the dummy warhead shall be interpreted as proper performance of the arming circuits, dummy timer and pressure check.

3.5.2 STATIC LOADS. The dummy warhead shall be capable of withstanding the static loads resulting from suspension, handling and loading.

3.5.3 SHOCK LOADS. The dummy warhead shall be capable of meeting the functional requirements of this specification after being subjected to the shock loads incident to handling, suspension and loading.

3.6 ENVIRONMENT. The dummy warhead shall meet the functional requirements during or after, as applicable, subjection to salt fog, likely to be encountered during transportation, storage, or use on or near a large body of salt water.

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3.7 IDENTIFICATION OF PRODUCT. Equipment, assemblies and parts shall be marked for identification in accordance with MIL-STD-130.

3.8 WORKMANSHIP. Workmanship requirements shall be as specified in MIL-B-27000.

4. QUALITY ASSURANCE PROVISIONS

4.1 CLASSIFICATION OF TESTS. The inspection and testing of the dummy warhead shall be classified as follows:

a. Preproduction Testing

b. Acceptance Tests

4.2 PREPRODUCTION TESTING

4.2.1 PREPRODUCTION TEST SAMPLES. The preproduction test samples shall consist of units representative of the production equipment, complete with lead ballast. They shall be tested at a laboratory designated by the procuring activity, or, when so stated in the contract, at the contractor's plant under the supervision of the procuring activity.

4.2.1.1 TEST REPORTS. In the event that preproduction testing is performed at the contractor's plant, the contractor shall, after completion of the preproduction tests, prepare a preproduction test report in accordance with MIL-T-9107, and furnish three complete copies of the report to the procuring activity.

4.2.2 PREPRODUCTION TESTS. Preproduction tests shall consist of all tests described under "4.5 TEST METHODS," and shall be arranged in the same order.

4.3 ACCEPTANCE TESTS. Acceptance tests shall consist of the following individual tests:

a. Examination of Product

b. Conformance to Drawings

c. Continuity Test

4.4 TEST CONDITIONS. The test conditions are described under the test methods to which they apply.

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4.5 TEST METHODS

4.5.1 EXAMINATION OF PRODUCT. Each dummy warhead shall be examined for strict conformance to the requirements of this specification for which tests are not specified. Failure to meet these requirements shall be cause for rejection.

4.5.2 CONFORMANCE TO DRAWINGS. Each dummy warhead shall be inspected for conformance to the drawings, specifications and standards specified therein. Failure to meet these criteria shall be cause for rejection.

4.5.3 CONTINUITY TEST. A complete continuity test shall be conducted on each dummy warhead to insure that all installations are properly wired and each unit will function as intended. The following conditions shall be met:

a. In the safe position the arming switch will be locked until 28 volts are applied to pins J and B of J-1 connector.

b. With the arming switch in the safe position there shall be continuity between pins E and N to pin L of J-1 connector.

c. With the arming switch in strike free fall or strike retarded position there shall be continuity between pins E and N to pin M of J-1 connector.

d. With the pressure tube full of air at 2 PSIG there shall be continuity between pin A and pin C of J-4 connector.

e. With the pressure tube full of air at 7 PSIG there shall be continuity between pin F and pin B, pin B and pin E of J-4 connector and between pin B of J-4 connector and pin K of J-2 connector.

f. There shall always be continuity between pin D of J-4 connector and pin E of J-3 connector, also between pin A of J-2 connector and pin D of J-3 connector.

Failure to produce the continuity required by this test shall be cause for rejection.

4.5.4 STATIC LOAD TEST. The static load test shall consist of combined fore and aft, side and vertical loads.

4.5.4.1 TEST SET UP

a. The complete dummy warhead shall be suspended by the spoiler ring at station 97.263 from a test frame which is capable of withstanding the applied loads. An additional fixture shall be attached

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to the forward end by means of the two uppermost 5/8 inch holes and the 5/8 inch hole on the center line of the dummy warhead. Two straps three inches wide shall be used as a safety measure; one of these straps shall be located at station 8.800, and the other shall be located at station 79.750. These straps shall be tightened to sufficiently prevent the dummy warhead from being displaced by failures in the suspension system.

b. Equipment capable of applying loads specified in the test procedures shall be used. An adequate load measuring device, such as a load-cell, shall be used in conjunction with the load applying equipment.

4.5.4.2 TEST PROCEDURE. The suspension test of the dummy warhead shall consist of the simultaneous imposition of vertical and fore and aft loads. The loads shall first be imposed simultaneously in steps of 0, 50, 66.7, 75, 85, 90 and 100 percent of design limit loads. After the ability of the dummy warhead to withstand 100 percent of the design limit loads for each of the three cases has been substantiated, the test shall be repeated using the same percent loading increments of 150 percent of the design limit loads. Any permanent deformation after loading to 115 percent of the design limit load or breakage while loading to 150 percent of design limit load shall constitute cause for rejection. Design limit loads are:

a. In the vertical plane:

5.0 G's up

5.0 G's down

b. In the horizontal plane:

7.5 G's forward

7.5 G's aft

4.5.5 SHOCK TEST

4.5.5.1 TEST SET UP

4.5.5.2 TEST SPECIMEN PREPARATION. The dummy warhead to be subjected to shock testing shall have the lead ballast installed.

4.5.5.3 TEST SET UP. The test set up shall be as specified for the static load test.

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4.5.5.4 TEST PROCEDURE. Test procedure shall be in accordance with MIL-B-27000. Immediately after subjection to the shock test, the dummy warhead shall be subjected to the continuity test. Failure or permanent deformation of any part or failure to pass the continuity test shall be cause for rejection.

4.5.6 SALT FOG TEST. The dummy warhead shall be subjected to the salt fog test described in MIL-E-4970.

5. PREPARATION FOR DELIVERY

5.1 LEVEL OF PRESERVATION AND PACKING

5.1.1 LEVEL A. Dummy Warhead, Bomb MDU-9/E shall be preserved in accordance with MIL-P-116, method 1A.

5.1.2 LEVELS B AND C. Dummy Warhead, Bomb MDU-9/E shall be preserved in accordance with the manufacturer's commercial practice.

5.2 LEVELS OF PACKING

5.2.1 LEVEL A. Dummy Warhead, Bomb MDU-9/E preserved to meet 5.1.1 shall be packaged in exterior type shipping containers meeting MIL-C-104, type II, class 1 or 2, style a. As far as practicable, exterior containers shall be of uniform shape and size, be of minimum cube and tare consistent with the protection required. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. Containers shall be provided with a case liner conforming to MIL-L-10547 and shall be sealed in accordance with the appendix thereto.

5.2.2 LEVEL B AND C. Dummy Warhead, Bomb MDU-9/E preserved to meet 5.1.2 shall be packed in a nailed assembly crate conforming to MIL-C-3774. As far as practicable, exterior containers shall be of uniform shape and size, of minimum cube and tare consistent with the protection required. The crate shall be closed and strapped in accordance with the applicable container specifications or appendix thereto.

5.3 PHYSICAL PROTECTION. Cushioning, blocking, bracing, bolting and testing to determine adequacy as required shall be in accordance with JAN-P-100.

5.4 MARKING. Interior and exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be:

DUMMY WARHEAD, BOMB MDU-9/E

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6. NOTES

6.1 INTENDED USE. The MDU-9/E, Dummy Warhead is intended to be used by ground crews to gain practice in ground handling and loading of the parent weapon. The item is non-expendable and is not normally dropped from the carrying aircraft.

6.2 ORDERING DATA. Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Whether preproduction tests will be conducted by the procuring activity.

6.2.1 PREPRODUCTION TEST SAMPLES. It is expected that the contract or purchase order will specify that two units representative of the production dummy warhead will be submitted to the procuring activity for preproduction test and evaluation. The invitation for bids and the contract or purchase order should specify the point of inspection for these tests.

6.3 PRODUCTION DELIVERY. No production items shall be delivered prior to approval of the preproduction samples.

6.4 DEFINITIONS

a. PREPRODUCTION TESTS. Preproduction tests are those tests performed prior to production on representative samples of the production equipment to demonstrate that the equipment meets the requirements of this specification.

b. ACCEPTANCE TESTS. Acceptance tests are those tests accomplished on representative samples of equipment submitted for acceptance under a contract to determine conformance to the requirements of this specification, and to determine that the equipment is equivalent in performance and construction to the approved preproduction samples.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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